

Improving Operational Efficiency in B2C Agribusiness: A Data-Driven Analysis for Sairanga Pesticides & Seeds

A Proposal report for the BDM Capstone Project

Submitted by

Name : Vaddi Yaswanth

Roll number :23f3002138



IITM Online BS Degree Program,
Indian Institute of Technology, Madras,
Chennai Tamil Nadu, India, 600036

Contents

1. Executive Summary	3
2. Organisation Background	3
3. Problem Statement	4
3.1 Optimizing Customer Segments Through Purchase Pattern Analysis	
3.2 Enhancing Inventory Control with Demand Forecasting	
3.3 Improving Profitability Through Product Wise Margin Analysis	
4. Background of the Problem	4
5. Problem Solving Approach	5
6. Expected Timeline	6
7. Expected Outcome	7

Declaration Statement

I am working on a Project Title “Improving Operational Efficiency in B2C Agribusiness: A Data-Driven Analysis for Sairanga Pesticides & Seeds”. I extend my appreciation to Sairanga Pesticides & Seeds, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project undertaken towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

V. Yaswanth

Name: Vaddi Yaswanth

Date: 12/10/2025

1. Executive Summary

This Business Data Management (BDM) Capstone Project is part of the BS in Data Science and Applications program offered by IIT Madras. The project focuses on Sairanga Pesticides & Seeds, a B2C agribusiness retail outlet located in Kanaganapalli, Sri Sathya Sai District, which has been serving local farmers since 2021. The shop primarily sells pesticides, fertilizers, and crop protection products, playing a key role in helping farmers maintain healthy crops and achieve higher yields.

Even though the shop has good product offerings, it is confronted with a number of business issues that hinder its growth. These are a lack of proper customer segmentation, poor inventory management, and varying profit margins among product lines. A majority of these problems are caused by the lack of a formal analysis of data and poor visibility into key business segments like sales trends, product movement, and customer buying patterns.

To address these challenges, a structured data driven strategy will be implemented using analytical and forecasting techniques. Customer Segmentation Analysis will be performed through RFM (Recency, Frequency, Monetary) analysis to classify customers based on purchase behavior, spending patterns, and seasonal trends. Demand Forecasting and Inventory Optimization will leverage Time Series Models such as ARIMA or Exponential Smoothing to predict future demand accurately and maintain optimal stock levels.

By applying these methods, Sairanga Pesticides & Seeds can strengthen its business in multiple ways. It will be able to make better pricing decisions, manage inventory more efficiently, and understand its customers more deeply. This approach will not only help increase profitability and customer retention but also foster a data based decision making culture, setting the foundation for sustainable growth in the future.

2. Organisation Background

Sairanga Pesticides & Seeds is a B2C agribusiness retail outlet located in Kanaganapalli, Sri Sathya Sai District. The shop primarily serves local farmers by providing essential agricultural inputs such as pesticides, fertilizers, plant growth promoters, and crop protection products. By offering reliable and affordable solutions, Sairanga Pesticides & Seeds supports farmers in maintaining healthy crops and improving overall yield and productivity.

With an average monthly sale of ₹2–3 lakh and a small team of 3 dedicated employees, the shop has built strong relationships with nearby farmers through consistent product availability and personalized guidance. Its operations are centered on trust, quality, and customer satisfaction, ensuring that every farmer receives the right product recommendations based on crop type and season.

Despite being located in a rural area, Sairanga Pesticides & Seeds continues to grow steadily by adopting modern business practices and expanding its product range to meet diverse farming needs. Its commitment to honest service, quality assurance, and farmer welfare has established it as a dependable and trusted agricultural input provider in the local community.

3. Problem Statement

3.1 Optimizing Customer Segments Through Purchase Pattern Analysis: The shop does not maintain structured records of customer behavior such as product preferences or purchase frequency. This limits the ability to design loyalty programs, seasonal discounts, or personalized offers.

3.2 Enhancing Inventory Control with Demand Forecasting: Unsystematic stocking practices cause overstocking of slow moving goods and stockouts of popular products, leading to lost sales and wastage from expired pesticides.

3.3 Improving Profitability Through Product Wise Margin Analysis: Lack of margin visibility across brands and product types makes it difficult to identify profitable items or adjust pricing effectively.

4. Background of the Problem

Sairanga Pesticides & Seeds, a retail agribusiness store in Kanaganapalli, Sri Sathya Sai District, operates in a competitive market where farmers buy fertilizers, pesticides, and crop protection products based on crop season and local demand. Although the shop keeps basic sales records, this data is not properly analyzed to understand customer needs or product trends. Because of this, most business decisions are taken without clear insights.

The main problems come from inside the business. There is no proper system to keep track of customer details or their buying history.

The shop also does not use any method to estimate which products will be in demand during different seasons. In addition, there is no clear idea of profit margins across different products and suppliers, making it hard to focus on the most profitable items.

Without using data in a structured way, the shop may continue to face problems like low profits, excess or insufficient stock, and losing customers to competitors. This project aims to use the available sales and customer data to build a simple, insight based system that can help the shop grow steadily and serve its customers better.

5. Problem Solving Approach

To solve Sairanga Pesticides & Seeds key operational challenges such as unclear customer segmentation, inefficient inventory management, and inconsistent product profitability a multi method analytical approach will be adopted. The core methods include Data Analysis, Predictive Modeling, and Performance Benchmarking.

5.1 Methods Used & Justification

Customer Segmentation Analysis will use RFM (Recency, Frequency, Monetary) modeling and clustering to group customers by their buying behavior, visit frequency, and spending. This will identify loyal, seasonal, and inactive customers. Association rule mining (Apriori algorithm) will find commonly bought product combinations, helping create targeted promotions and crop specific bundles. These methods will enable Sairanga Pesticides & Seeds to personalize marketing, boost engagement, and encourage repeat purchases.

Inventory Optimization and Demand Forecasting will use Time Series Analysis and ABC Classification. Models like ARIMA or Prophet will analyze past sales to identify seasonal demand patterns, such as Kharif and Rabi crops, helping forecast future sales and maintain proper stock levels.

ABC Classification will categorize products into high, medium, and low priority based on their sales value. These methods will reduce stockouts, minimize expired inventory, and optimize working capital.

Product Wise Profitability Analysis will use Gross Margin and Contribution Margin calculations, applying the formula $(\text{Profit} = \text{Selling Price} - \text{Cost of Goods})$ Sold for each product to identify the most profitable pesticides and fertilizers. A Pareto (80/20) Analysis will determine the top 20% of products generating most of the profits.

Additionally, Scenario Modeling will simulate pricing and cost changes, helping management evaluate the financial impact of different strategies. This approach ensures Sairanga Pesticides & Seeds focuses on high margin products and supports sustainable growth.

5.2 Data Collection: Strategy and Relevance: To address the identified business challenges effectively, a focused and comprehensive dataset will be collected from operational and transactional sources. This includes customer level data such as purchase frequency, product preferences, crop type, and total spending to assess buying behavior and segment customers effectively. Inventory level data such as monthly stock levels, supplier lead times, reorder frequency, and expiration dates will help optimize stock and reduce wastage. Product level data including cost, selling price, units sold, and category will provide insights into profitability and margin trends.

5.3 Analysis Tools: A focused set of tools will be used for analysis. Google Sheets or Excel will be the main platforms for cleaning, transformation, and exploratory analysis using formulas, pivot tables, and charts, enabling detailed examination of customer behavior, product performance, and inventory trends. Python library Pandas will be used selectively for advanced calculations and automation tasks where needed. Power BI will provide interactive dashboards to visualize trends and insights effectively. These tools are chosen for their accessibility, analytical power, visualization strength, and relevance for handling small to medium business datasets efficiently.

6. Expected Timeline

6.1 Work Breakdown Structure:

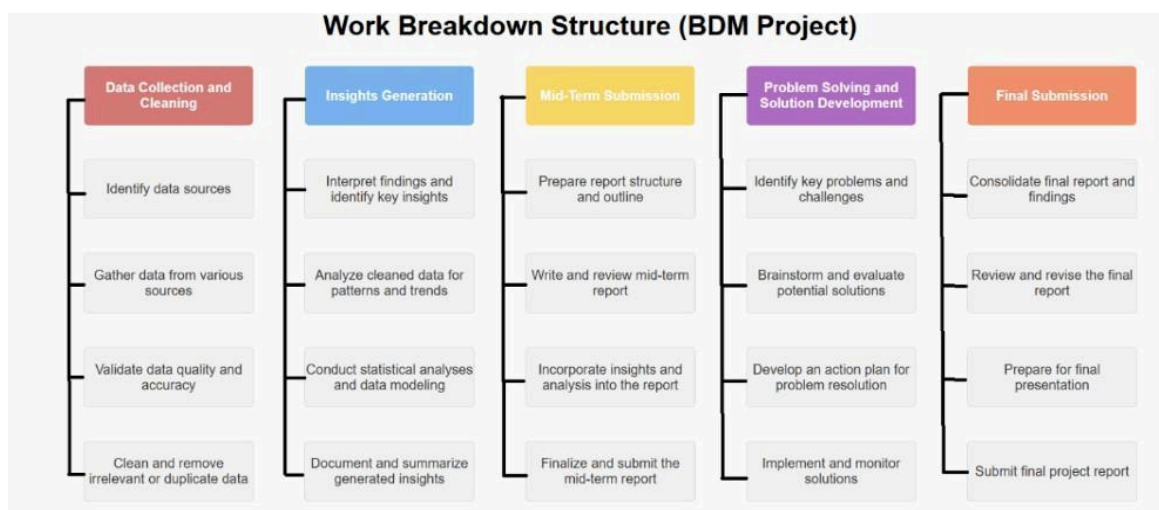


Figure 1: Work Breakdown Structure

6.2 Gantt Chart:

Project Timeline

Task	Start Date	End Date
Meeting 1	01/09/2025	03/09/2025
Meeting 2	05/09/2025	07/09/2025
Data Collection Round 1	16/09/2025	28/09/2025
Proposal Submission	17/10/2025	19/10/2025
Data Collection Round 2	18/10/2025	22/10/2025
Data Pre-Processing	25/10/2025	28/10/2025
Data Analysis	06/11/2025	08/11/2025
Mid Term Submission	15/11/2025	30/11/2025
Prepare Final Report	01/12/2025	03/12/2025
Final Presentation Submission	01/12/2025	03/12/2025



Figure 2: Project timeline Gantt Chart

7. Expected Outcome

7.1 Optimized Customer Segmentation: By analyzing purchase patterns using RFM and clustering, the shop will gain structured insights into customer behavior. This will enable targeted loyalty programs, seasonal promotions, and personalized offers, improving customer engagement and repeat purchases.

7.2 Improved Inventory Management: Predictive demand forecasting will help maintain optimal stock levels, reducing overstocking of slow moving items and preventing stockouts of popular products. This will lower wastage from expired goods and improve inventory turnover.

7.3 Enhanced Product Profitability: By analyzing product wise margins, the shop will identify high margin items and make informed pricing and sourcing decisions. This will maximize profitability and ensure focus on the most valuable products.